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classmate

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$$\left. \begin{array}{l} \{T_n\}, \theta \\ \sqrt{n}(T_n - \theta) \xrightarrow{D} N(0, \sigma^2) \end{array} \right\} \text{(given)}$$

* Let $\{T_n\}$ be a seq. of est. for given parameter θ .
 For a +ve seq. $\{b_n\}$ s.t. $b_n \xrightarrow{n \rightarrow \infty} \infty$, we say T_n
 is b_n -consistent if $b_n(T_n - \theta) = O_p(1)$
 $\Rightarrow T_n - \theta = O_p(\frac{1}{b_n})$
 (bounded in probability)

$\rightarrow F(\theta) = \frac{1}{2}$. F is strictly inc. and cont. in a nbd of θ .

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$

$\tilde{X}_n = \text{sample median}$.

In order to show the consistency for $\tilde{X}_n$ an est. for θ , we need to:

$$P(\tilde{X}_n - \theta > \varepsilon) \xrightarrow{n \rightarrow \infty} 0 \quad \text{-- (i)}$$

$$P(\tilde{X}_n - \theta < -\varepsilon) \xrightarrow{n \rightarrow \infty} 0 \quad \text{-- (ii)}$$

(i) is same as $P(\text{at least half of the } X_i\text{'s are } > \theta + \varepsilon)$
 $\rightarrow$ Binomial prob., i.e., $\text{Bin}(n, 1 - F(\theta + \varepsilon))$

(ii) is same as $P(\text{at least half of the } X_i\text{'s are } < \theta - \varepsilon)$
 $\rightarrow$ Binomial prob., i.e., $\text{Bin}(n, F(\theta - \varepsilon))$

In both cases $1 - F(\theta + \varepsilon) < \frac{1}{2}$ & $F(\theta - \varepsilon) < \frac{1}{2}$, $\therefore$ The prob. will go to 0.

- Note: For any problem involving Cauchy distribution, starting with sample median is good.

* Simple linear regression model

$$Y_i = \alpha + \beta X_i + \varepsilon_i \quad \varepsilon_1, \dots, \varepsilon_n \text{ iid } F$$

$$\hookrightarrow \text{const.} \quad E(\varepsilon_i) = 0, \text{Var}(\varepsilon_i) = \sigma^2$$

α, β are unknown parameters, X_i 's are constants.

$$\hat{\beta}_{LS} = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n Y_i (X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{\alpha}_{LS} = \bar{Y} - \hat{\beta}_{LS} \bar{X}$$

$$E(\hat{\beta}_{LS}) = \beta \quad \text{unbiasedness.}$$

$$\text{Var}(\hat{\beta}_{LS}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \rightarrow 0 \quad \text{if} \quad \sum_{i=1}^n (X_i - \bar{X})^2 \rightarrow \infty$$

$\therefore$ For consistency, X_i 's must be sufficiently spread out.

$$E(\hat{\alpha}_{LS}) = \alpha$$

Work out $\hat{\alpha}$'s consistency. Assume $\bar{X}$ converges to a const.

* $\{X_n\}$, $X : X_n \xrightarrow{D} F$

$$\{F_n\}, \quad F : F_n \xrightarrow{L} F \quad \text{(only on cont. pts. of } F \text{)} \\ \text{(same thing)}$$

Q) $X_1, \dots, X_n \text{ iid } f(x|\theta)$ (pdf / pmf)

$$\hat{\theta}_n = \text{MLE for } \theta$$

$$= \arg \max_{\theta} \prod_{i=1}^n f(X_i | \theta)$$

In many cases, $\hat{\theta}_n$ will be consistent.
(Binomial, Poisson, Exponential, Normal)

→ $X_1, \dots, X_n$ iid $\text{Unif}(0, \theta)$

In this case $(\hat{\theta}_n)_{\text{MLE}} = X_{(n)}$,

Q) Is it consistent?

A) $E(X_{(n)}) = \frac{n\theta}{n+1}$

We can easily see that $P(\hat{\theta}_{\text{MLE}} < \theta - \varepsilon) \rightarrow 0$. (I just used cdf)
And $P(\hat{\theta}_{\text{MLE}} > \theta + \varepsilon) = 0$ always.

Q) Show consistency of $\hat{\theta}_{\text{MLE}}$.

Exercise

A) Yes, if regularity conditions on $F(X, \theta)$ hold & ML eqⁿ has unique solⁿ for each n .